

MemeClassroom

Educational Meme Platform

Project Framework & Feature Documentation

Version 1.0 | Ready for Development

June 2026

1. Project Overview

MemeClassroom is a web-based educational platform that bridges the gap between humour and learning. Teachers, students, and subject-matter experts collaborate on meme-based content to make topics more engaging, memorable, and enjoyable for students of all ages.

The platform is built around five core pages — Meme Lab, Meme Library, Meme Resources, Staffroom, and an Admin Panel — each serving a distinct purpose in the ecosystem. A sixth section, User Profile, ties individual contributions together and recognises platform engagement through a badge system.

Core value proposition

- Students retain information longer when it is paired with humour and visual context.
- Teachers need a trusted, curated space to find accurate and age-appropriate memes.
- Subject experts provide credibility through a verified comment system.
- The Staffroom enables real-world feedback loops: teachers share what actually worked in class.

Design principles

- Mobile-first, responsive design for classroom accessibility.
- Social-media familiarity (feed, likes, saves) to reduce the learning curve.
- Role-based access control ensures content quality and community safety.
- Modular architecture so features can be shipped incrementally.

3. User Roles & Permissions

MemeClassroom operates on a three-tier role system. Roles are assigned at registration (Teacher/Student) or through an admin-verified application process (Expert). Admins are internally appointed.

Permission	Student	Teacher	Expert	Admin	Guest
View Meme Library and resource page	Yes	Yes	Yes	Yes	Yes (limited)
Create memes (Lab)	Yes	Yes	Yes	Yes	Limited (2)
Post to Library	Yes	Yes	Yes	Yes	No
Rate / like memes	Yes	Yes	Yes	Yes	No
Post in Staffroom	Yes	Yes	Yes	Yes	No
Expert comment section	No	No	Yes	Yes	No
Upload to Resources	Yes	Yes	Yes	Yes	No
Apply for Expert role	Yes	Yes	N/A	N/A	No
Approve Expert apps	No	No	No	Yes	No
Ban / moderate users	No	No	No	Yes	No
View admin dashboard	No	No	No	Yes	No

4. Homepage

The homepage is the public-facing landing page of MemeClassroom. It serves as the entry point for both new visitors and returning users. Its primary job is to communicate the platform's value instantly and direct users toward the relevant section.

Purpose

- Introduce MemeClassroom and explain its core concept to first-time visitors.
- Display platform statistics (memes created, teachers active, subjects covered) to build trust.
- Showcase trending memes from the library to demonstrate content quality.
- Present clear calls-to-action: Browse Library and Create a Meme.
- Explain the three user roles — Teacher/Student, Expert, Admin — so new users self-select.

Key UI sections

Section	Description
Hero banner	Full-width headline, tagline, and three CTA buttons (Browse Library / Create a Meme/Browse Resources). Includes a subject-tag badge pill.
Stats bar	Live counters: total memes, registered teachers, verified experts, subjects covered. Pulled from the database.
Feature cards	Four cards linking to Meme Lab, Library, Resources, and Staffroom with icon, title, short description, and a category chip.
Trending memes	Horizontally scrollable or grid row of the top 6 memes this week, ordered by aggregate rating score.
Roles section	Three role cards (Teacher/Student, Expert, Admin) explaining access level and how to apply for Expert status.
Top navigation	Logo, nav links to all pages, Sign In, and Join Free buttons. Avatar and notification bell for logged-in users.
Testimonials	User experiences
Donate	Donate (eg:paypal)
Advertisement	Space for Google/sponsored ads

Connected pages

The homepage links to every major section. The hero CTA buttons route to the Library, Resources and Meme Lab. The feature cards and trending memes each deep-link to their respective pages. The Roles section links to the registration or Expert application flow.

5. Meme Lab — Editor

Meme Lab is the creative workspace of MemeClassroom. It is a full-featured meme editor that supports multiple media types — images, videos, audio clips, and GIFs — with a rich set of editing tools for text, cropping, colour adjustment, and sticker overlay. Any registered user (Teacher, Student, or Expert) can access the Lab to create and publish memes to the Library.

Purpose

- Provide a browser-based editor so users never need third-party tools.
- Support the four primary meme formats: image, video, GIF, and audio-based.
- Allow users to add text overlays, adjust styling, and choose stickers.
- Let users tag memes by subject, topic (text), grade, format, and language before publishing.
- Auto-save drafts so work is never lost between sessions.

Editor toolbar — media type tabs

Tab	Supported input	Notes
Image	JPEG, PNG, WebP, SVG	Primary format. Drag-and-drop or file browser.
Video	MP4, MOV, WebM (max 50 MB)	Trimming and text overlay on video frames.
Audio	MP3, WAV, OGG (max 20 MB)	Waveform visualiser; captions can be added.
GIF	GIF upload or Giphy API search	Search Giphy directly inside the editor.

Editing tools

Tool	Function
Text overlay	Add top and bottom text with font, size, colour, and shadow controls. Drag to reposition.
Crop	Free-form or aspect-ratio-locked cropping (1:1, 16:9, 4:3).
Brightness / contrast	Slider-based adjustment for image and video frames.
Stickers	A library of educational and reaction stickers. Users can also upload custom stickers.
Palette	Colour picker for text and shape overlay fill.
Typography	Font family selector (6 built-in options including Impact, Arial, Comic Sans equivalents).
Templates	Admin added tools + Upload option for users

Tutorial	Guided tutorial/guidelines on meme creation
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Publish metadata form

Before publishing, the user fills in a metadata form below the canvas. This data powers filtering and discovery in the Meme Library.

Field	Options	Used for
Subject / topic	Biology, Physics, Maths, Chemistry, History, Geography, etc.	Library filters and expert routing.
Grade	Ages 10-12, 13-15, 16-18, University	Safe content filtering and recommendations.
Format	Image, Video, GIF, Audio	Format-specific library filters.
Language	English, Hindi, Malayalam, Tamil	Filtering by language
Remarks	Any additional tags	Additional metedata

Connected pages

Meme Lab connects to the Meme Library on publish. Published memes appear immediately in the Library feed and are flagged for optional expert review if the subject matches an available verified expert. Draft memes are accessible from the user Profile page.

6. Meme Library

The Meme Library is the social heart of MemeClassroom. It functions like an educational social media feed where any registered user can browse, rate, save, and share memes. Verified experts can leave an expert comment on any meme within their subject area. The Library uses a multi-dimensional rating system rather than a simple like count, making it easier for teachers to identify the most pedagogically sound content.

Purpose

- Serve as the central discovery hub for all published memes on the platform.
- Allow multi-dimensional rating so teachers can judge accuracy, humour, and clarity independently.
- Provide a verified Expert comment section to give memes academic credibility.
- Enable filtering by subject, age group, format, and quality markers (Expert verified, Top rated).
- Allow users to save memes to their personal collection and share links externally.

Feed layout & filtering

Feature	Detail
Filter chips	Subject (Biology, Maths, Physics, etc.), Grade, Language, Format, Expert verified, Top rated. Multiple filters can be active simultaneously.
Sort options	Trending (composite rating), Newest, Most saved, Most liked, Expert picks.
Meme card	Shows thumbnail, subject tag, title, age group, rating bars, who contributed, Expert badge if reviewed, like count, edit/customise button, save button, and share button.
Rating bars	Four bars per meme: Age appropriateness, Language appropriateness, Content Validity, and Creativity . Each bar shows the community average out of 5 from all user ratings.
Expert badge	A green shield icon appears on any meme that has at least one verified Expert comment.
Infinite scroll	Feed loads 20 memes at a time; more load on scroll. Server-side pagination via cursor-based API.

Rating system

When a user opens a meme, they can submit their own rating across four dimensions. Each dimension is scored from 1 to 5. The meme card in the feed shows the community average for each dimension as a filled progress bar.

- Language appropriateness -
- Age- appropriateness
- Content- validity
- Creativity

Expert comment section

Each meme has one comment area: and an Expert Comments section exclusive to users with the Verified Expert role. This separation ensures that expert insights (e.g. 'This accurately represents semi-conservative replication') are visually distinct from general user feedback.

Comment type	Who can post
Expert comments	Verified Experts only. Displayed in a distinct, highlighted box with expert's name, field, and institution.

Connected pages

The Library connects to the Meme Lab , the Staffroom (a meme card can be tagged when posting a classroom story), and the User Profile (saving a meme adds it to the user's collection). Admin moderation tools can flag or remove any meme from the Library view.

7. Meme Resources

Meme Resources is the knowledge base of MemeClassroom. It hosts articles, research papers, videos, stories, and downloadable activity packs — all centred on meme-based learning. This section gives the platform academic depth and helps teachers justify meme use to school leadership.

Purpose

- Provide evidence-based resources that validate meme-based pedagogy.
- Host lesson plan templates and classroom activity packs teachers can download directly.
- Feature storytelling content (e.g. 'How I used meme in my maths classroom').
- Allow Teachers and Experts to submit their own articles and videos for review.

Resource types

Type	Format	Who can submit
Article	Written post (rich text)	Teachers, Students, Experts (reviewed by Admin before publishing)
Research paper	PDF upload with abstract	Experts only (fast-tracked for publishing)
Video	Embedded YouTube / Vimeo link	Teachers, Experts
Meme Story	Stories behind memes	Experts
Activity	PDF or DOCX download, Embedded YouTube / Vimeo link	Teachers, Experts

Connected pages

Resources connects to the Admin Panel for submission approval.

8. Discussion / Staffroom

The Staffroom is MemeClassroom's community forum. It is modelled after a real staffroom conversation where teachers and students share what happened when they actually used a meme in class. Each post is linked to a specific meme from the Library and contains a narrative experience report. Other users can add comments or tips in a threaded reply system.

Purpose

- Enable teachers to share honest, real-world results from using memes in lessons.
- Build a feedback loop so meme creators know what works and what does not.
- Allow peer support: teachers replying to each other with extra tips and advice.
- Surface the best classroom experiences to the homepage 'Trending' section.

Post structure

Field	Detail
Author	User avatar, name, role badge (Teacher / Student), and posting date.
Linked meme OR upload images/embedd video	A tag chip linking to the specific meme used in class. Required field.
Class context	Optional: subject, year group, and approximate class size.
Experience body	Free-text field where the user describes what happened (min 50 characters to encourage quality posts).
Thread replies	Nested comment system. Any registered user can reply. Replies show avatar, name, and timestamp.
Likes	Users can like a post. High-liked posts surface on the homepage.

Connected pages

The Profile page shows all of a user's Staffroom posts.

9. Admin Panel

The Admin Panel is the control centre of MemeClassroom, accessible only to users with the Admin role. It covers four main areas: platform analytics, user management, Expert verification, and content moderation. Admins are appointed internally and cannot self-assign the role.

Purpose

- Monitor platform health through key analytics metrics.
- Manage all user accounts: view, edit, suspend, or permanently ban users.
- Approve or reject Expert role applications with credential review.
- Review and act on flagged content from the Library, Staffroom, or Resources.

Dashboard — analytics overview

Metric card	Data shown
Total users	Count of all registered accounts, broken down by role.
Memes published	Total meme count, with weekly/monthly growth trend.
Flagged content	Number of items currently pending moderation review.
Expert applications	Number of pending, approved, and rejected applications this month.
Active sessions	Real-time count of users currently on the platform.
Top subjects	Bar chart of the 5 most-memed subjects.

User management

The user table lists all accounts with name, email, role, meme count, join date, and current status. Admins can take the following actions:

Action	Effect
View profile	Opens a read-only view of the user's profile, posts, and meme history.
Edit role	Change a user's role between Student and Teacher (Expert requires formal approval flow).
Suspend	Temporarily disables login. User receives an email notification with reason.
Permanent ban	Deletes login access. Content is retained but attributed to [Deleted User]. Irreversible.
Send warning	Sends an in-app and email warning without restricting access.

Expert verification flow

When a user applies for Expert status, they submit their credentials (institution, field, qualification, proof document). The Admin reviews this submission and either approves (granting the Expert role immediately) or rejects with a reason message sent to the applicant.

Content moderation

Any registered user can flag a meme, Staffroom post, or Resource for review. Flagged items appear in the Admin moderation queue with the flagging reason and the number of users who flagged it. Admins can: approve the content (remove the flag), edit metadata, issue a warning to the creator, or remove the content entirely.

Connected pages

The Admin Panel connects to every other page in the platform. Admins can deep-link from the user table to a user's Profile, from flagged content to the specific Library or Staffroom item, and from Expert applications to the applicant's submitted credentials.

10. User Profile

Every registered user has a Profile page that acts as their personal dashboard on MemeClassroom. It displays their identity, role badges, contribution statistics, earned achievement badges, created memes, saved memes, and Staffroom posts. Experts also display their verified field and institution here.

Purpose

- Give users a personal identity and sense of ownership on the platform.
- Track and display contributions (memes created, likes received, staffroom posts).
- Reward engagement through a badge system.
- Let users manage their draft memes and saved collection.

Profile sections

Section	Content
Header	Avatar (initials or uploaded photo), display name, role badge(s), and an Edit Profile button for own profile.
Stats row	Three metric cards: memes created, total likes received, staffroom posts published.
Badges	Grid of earned achievement badges with icon, name, and tooltip description.
My memes	Grid of all published memes by this user. Clicking opens the meme in the Library.
Saved memes	Grid of memes the user has saved from the Library.
Staffroom posts	List of the user's discussion posts with linked meme and engagement count.
Drafts	Memos saved but not yet published. Only visible to the profile owner and admins.

Badge system

Badge	How to earn	Colour
Verified Expert	Granted by Admin on Expert role approval.	Purple
Top Creator	Publish 10+ memes with an average rating above 4.5.	Amber
Early Adopter	Register within the first 3 months of platform launch.	Teal
Community Voice	Post 10+ Staffroom discussions with 5+ replies each.	Blue

Trending Meme	Have a meme reach the homepage trending section.	Red
100 Saves	Have any single meme saved 100+ times.	Grey

11. Page Connections & Navigation Map

The table below shows how each page of MemeClassroom connects to the others. These connections define the navigation routes that need to be coded into the routing layer.

From page	Links / routes to
Homepage	Meme Lab (CTA), Library (CTA + trending cards), Resources (feature card), Staffroom (feature card), Sign In / Register.
Meme Lab	Library (on publish), Profile (save draft), Homepage (logo / nav).
Library	Meme Lab (Edit own meme), Staffroom (tag a meme from library), Profile (author avatar), Admin (flag content).
Resources	Library (linked meme references), Admin (submission approval), Profile (author credit).
Staffroom	Library (linked meme chip), Profile (author avatar + replies), Admin (flag post).
Admin	All pages (moderation deep-links), Profile (user detail view), Library / Staffroom (flagged content).
Profile	Library (meme grid items), Staffroom (post list), Meme Lab (edit draft), Admin (Expert application status).

12. Database Schema Overview

The following tables form the core relational schema for MemeClassroom. Each table lists its key fields. Foreign key relationships are noted where relevant.

Table	Key fields	Notes
users	id, name, email, password_hash, role, is_verified, created_at, banned	role ENUM: student, teacher, expert, admin
memes	id, creator_id (FK users), title, subject, age_group, format, visibility, media_url, created_at	Stores Cloudinary/S3 URL
ratings	id, meme_id (FK), user_id (FK), accuracy, humour, clarity	One rating row per user per meme
expert_comments	id, meme_id (FK), expert_id (FK), body, created_at	expert_id must have role=expert
general_comments	id, parent_id (self-FK), meme_id (FK), user_id (FK), body, created_at	Self-FK enables threading
resources	id, author_id (FK), type, title, body, file_url, status, created_at	status ENUM: pending, approved, rejected
staffroom_posts	id, author_id (FK), meme_id (FK), body, outcome_tag, likes, created_at	outcome_tag ENUM: worked, tweak, miss
staffroom_replies	id, post_id (FK), author_id (FK), body, created_at	Flat replies (not nested)
saves	id, user_id (FK), meme_id (FK), created_at	Junction table for saved memes
flags	id, reporter_id (FK), content_type, content_id, reason, status	content_type ENUM: meme, post, resource
badges	id, user_id (FK), badge_type, awarded_at	badge_type ENUM matches badge list
expert_apps	id, user_id (FK), field, institution, proof_url, status, reviewed_by	status ENUM: pending, approved, rejected

13. Development Checklist

Use this checklist to track feature completion during the build phase. Items are ordered by priority and dependency.

Phase 1 — Foundation

- Set up project repository (monorepo: /client + /server).
- Configure PostgreSQL database and run initial schema migrations.
- Implement user authentication: registration, login, JWT, OAuth (Google).
- Build role system: Student/Teacher on registration; Expert and Admin via admin assignment.
- Set up Cloudinary or AWS S3 media bucket with upload endpoint.
- Build responsive navigation bar with role-aware links.

Phase 2 — Core pages

- Homepage: hero, stats bar (live DB counts), feature cards, trending memes feed, roles section.
- Meme Library: feed with pagination, filter chips, sort options, meme cards with rating bars.
- Meme Library: rating submission (Accuracy / Humour / Clarity) per user per meme.
- Meme Library: General comment thread (nested replies).
- Meme Library: Expert comment section — visible only, not interactive, for non-experts.
- User Profile: stats row, my memes grid, saved memes, Staffroom posts list.

Phase 3 — Editor & Staffroom

- Meme Lab: canvas editor with image upload, top/bottom text overlay, font/colour controls.
- Meme Lab: GIF tab with Giphy API integration and search.
- Meme Lab: Video and audio upload with trimming and caption support.
- Meme Lab: Publish metadata form (subject, age group, format, visibility).
- Meme Lab: Draft auto-save to database.
- Staffroom: Post creation with required linked meme, free-text body, outcome tag.
- Staffroom: Threaded replies and post likes.

Phase 4 — Admin & Moderation

- Admin Panel: Analytics dashboard with key metric cards and subject bar chart.
- Admin Panel: User management table with suspend / ban / edit role actions.
- Admin Panel: Expert application review flow with credential document viewer.
- Admin Panel: Flagged content moderation queue with approve / remove actions.
- Flag system: flag button on memes, posts, and resources for all registered users.

Phase 5 — Resources & Badges

- Meme Resources: Resource feed with type filters; article and video viewer.
- Meme Resources: Submission form for Teachers and Experts; admin approval workflow.

- Badge system: automatic badge award triggers based on user activity events.
- Notifications: in-app notification bell for likes, replies, expert comments, badge awards.
- Email notifications: send on ban, warning, Expert approval, and new badge.
- Performance: add Redis caching for Library feed and homepage trending queries.